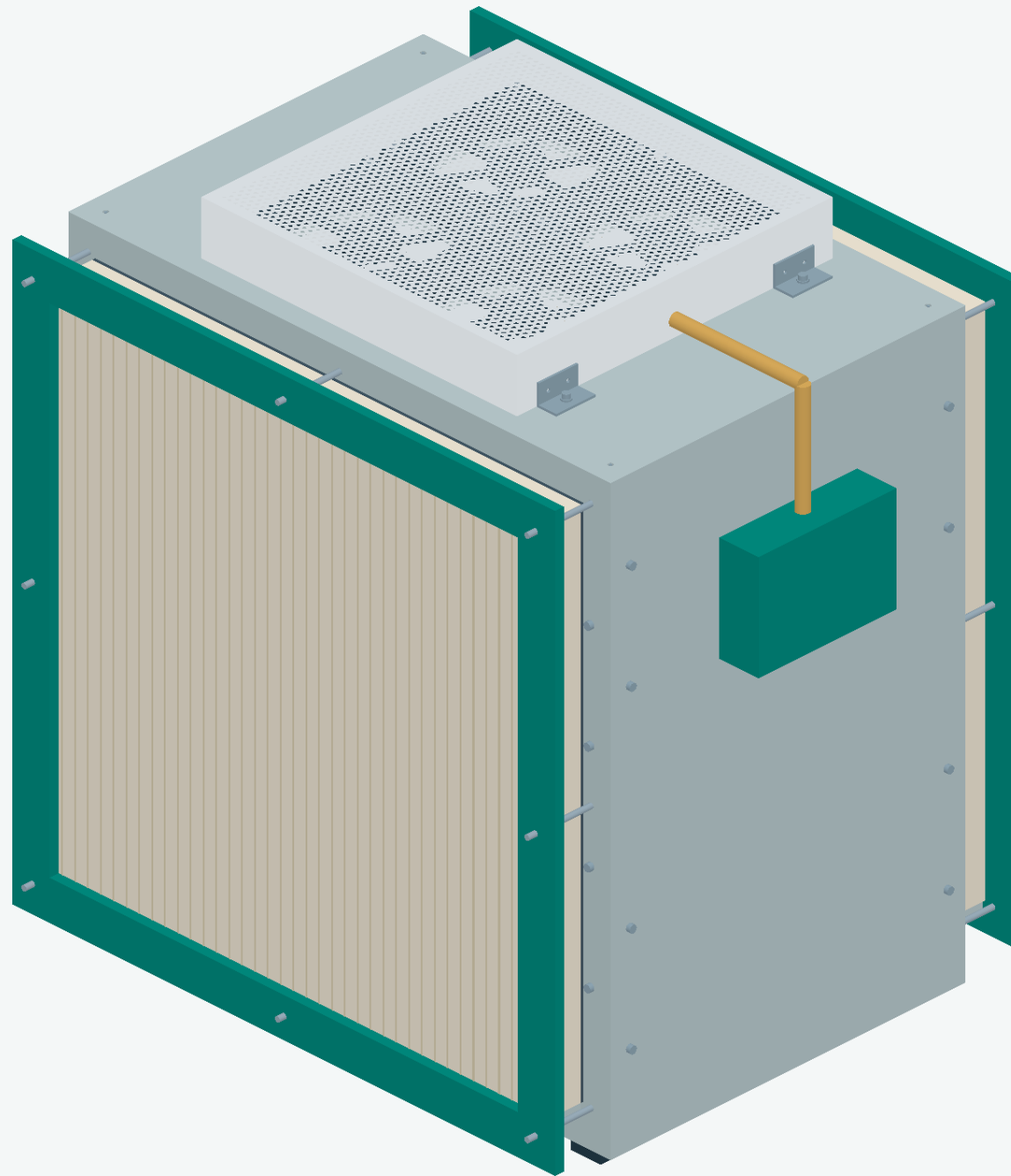


The prototype now has attachment details

R01 continues D01. It preserves the fan/filter arrangement and adds joints, support hardware and a cable-routing reservation.



Guard attachment

Eight angle brackets and four shared through-bolts connect both guards to the fan plate. Rivet-hole patterns are drilled in the actual CAD.

Cabinet joints

Thirty-two M4 through-bolt locations connect the end and filter-seat panels to the corner cleats. Top/base retention uses eight proposed timber screws.

Filter support

Eight metal angles support the four ledges. Shelf hardware is kept below the filter seating surface; flush screw-head detailing remains required.

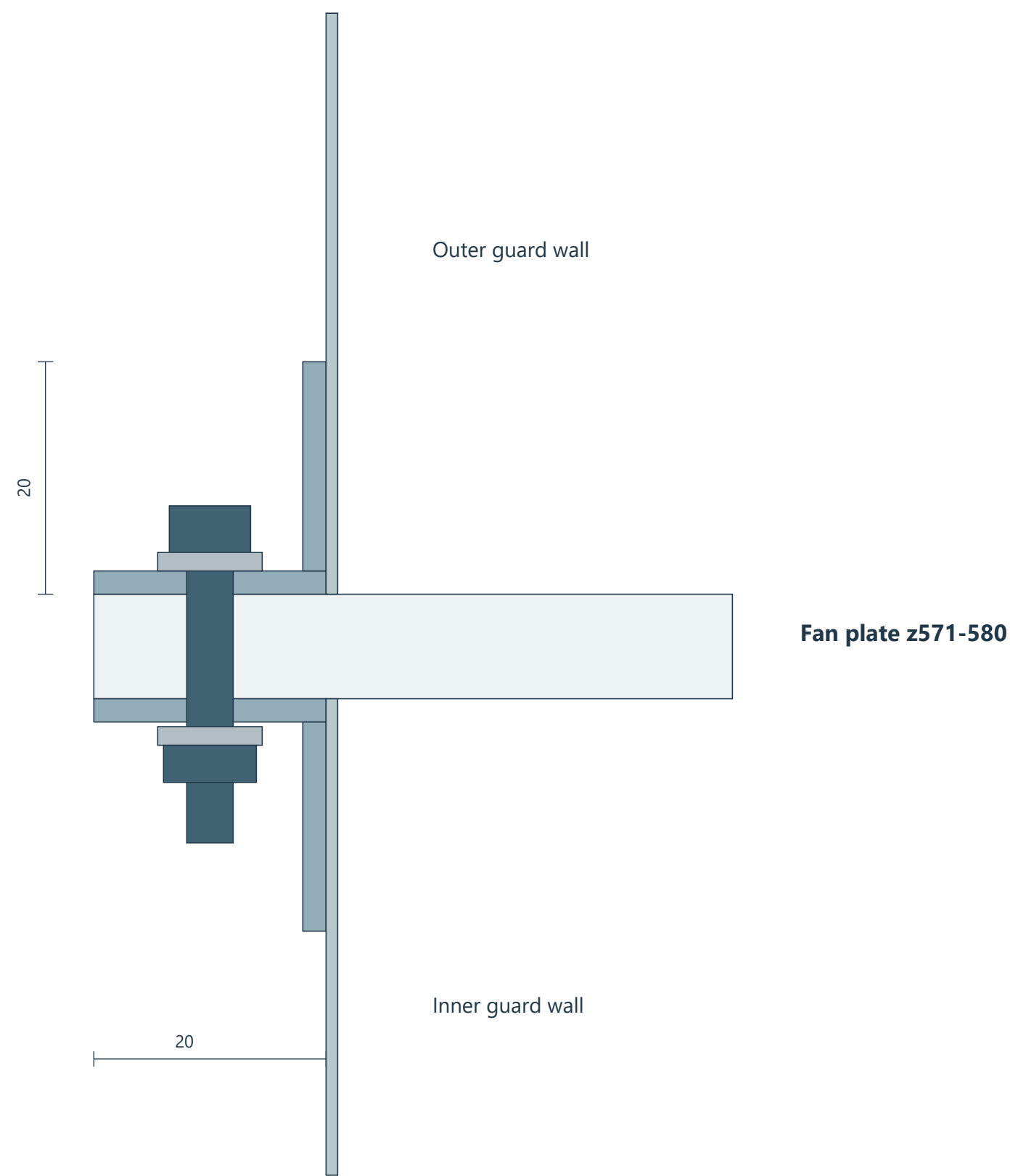
Cable routing

An external route is reserved from the right guard wall to the control box. It is not an approved cable-gland hole or completed electrical circuit.

CHECKED: 163 valid objects and zero nominal overlaps in 12,675 tested pairs. Reservations and inherited guard seams are excluded. These checks do not establish strength, finger protection, sealing or physical performance.

G03 Guard bracket and bolt section

Dimensions mm. Proposed angle material/grade, rivet grip and exact fasteners remain unselected. All dimensions are nominal.



Feature	Dimension / location
Each bracket	20 x20 x2 equal angle;40 long. Eight pieces. Nominal sharp corner; actual root fillet must fit.
Guard bolt pattern	Four dia4.5 holes through fan plate: x105/445 and y60/300. M4x25 proposed with two washers and a locknut.
Bracket foot hole	10 from free transverse edge;20 from length end. Hole dia4.5.
Skirt rivets	Two dia3.5 holes per bracket:10 and30 from length end. Outer z590; inner z560. Sixteen proposed3.2 rivets; grip stack3mm nominal.
Guard face	320 square with10 solid rim;300 square perforated field.4mm holes /6mm triangular pitch proposal. No holes cut into CAD sheet envelope.
Cage seams	Continuous closed joints of face-to-skirt and skirt corners, by shop-approved process for selected metal. No loose edges. Actual weld/formed joint design not released.

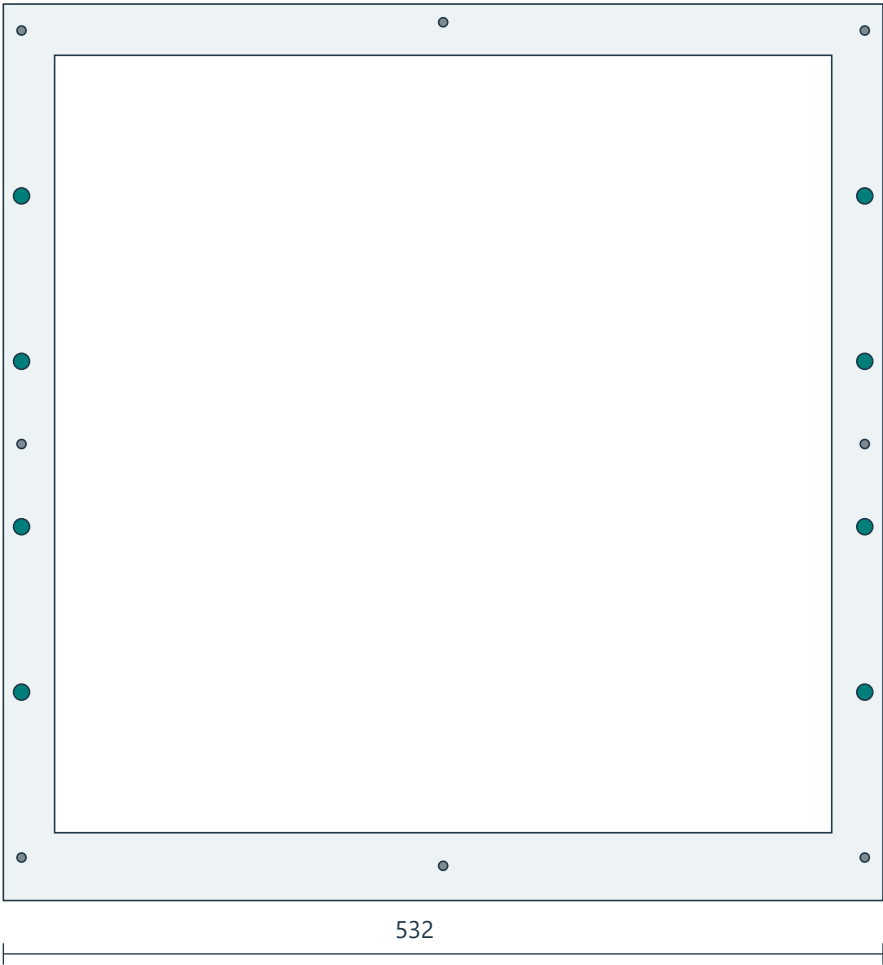
The lower washer has0.4mm nominal take-up in this envelope. Select the real washers and locknut, then verify engagement and tool clearance. The shared bolts mean removing the top guard can loosen the lower guard: support both and isolate power before service. No independent-access safety claim.

J01 Cabinet drilling and retention

Assembly coordinates: x from left; y from front seat; z from floor. No holes may be located by scaling this PDF.

Joint	Count / proposed hardware	Coordinates and assembly detail
End-to-cleat through joints	16 / M4x40 +washers/locknuts	Each end: y19/341; z105/205/405/505 End local datum y0,z29: vertical coordinates76/176/376/476. Drill dia4.5 through panel and cleat.
Seat-to-cleat through joints	16 / M4x40 +washers/locknuts	Each seat: x20/530; z155/255/355/455 Seat local datum x9,z29: x11/521,z126/226/326/426. Avoid filter clamp holes.
Top and base retention	8 / 4 top wood screws4x20;4 base4x25	x19/531; y19/341 Pilot diameter per actual wood/screw. Proposed seam: external removable foil HVAC tape after dry fit; exact tape unknown. Not a lifting structure.

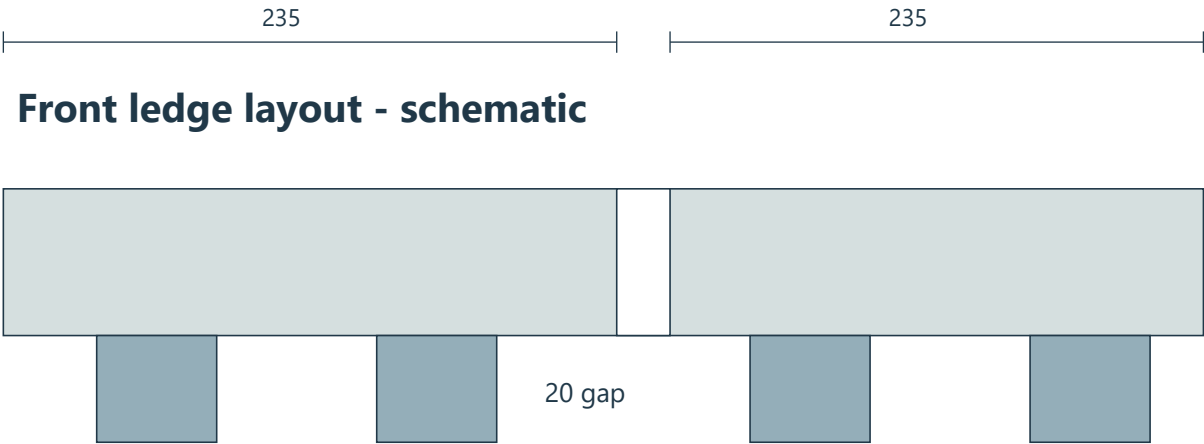
Seat panel - green = new cabinet joints



- Assembly order
- Dry-fit the base, panels and cleats square. Drill the through-joints using the panel datums; deburr and protect the edges. Fit washers and locking nuts on the internal side. Confirm the cabinet diagonals before tightening.
- Top and bottom
- Top screws4 x20 and base screws4 x25 are layout proposals, not a verified timber connection. Pilot diameter, screw type and end-grain holding strength must follow the selected wood and fastener data. No handles or lifting points are released: support the base when moving.
- Sealing
- Use compatible cured sealant on permanent seams. Proposed removable top seam uses externally applied, removable foil HVAC tape; product and adhesion are unverified. No new top gasket thickness is inserted silently into the CAD. Seal all penetrations and physically check for bypass.

M09 Filter ledge support

Each filter rests on two235mm ledges. Two metal angles support each ledge: eight brackets total.



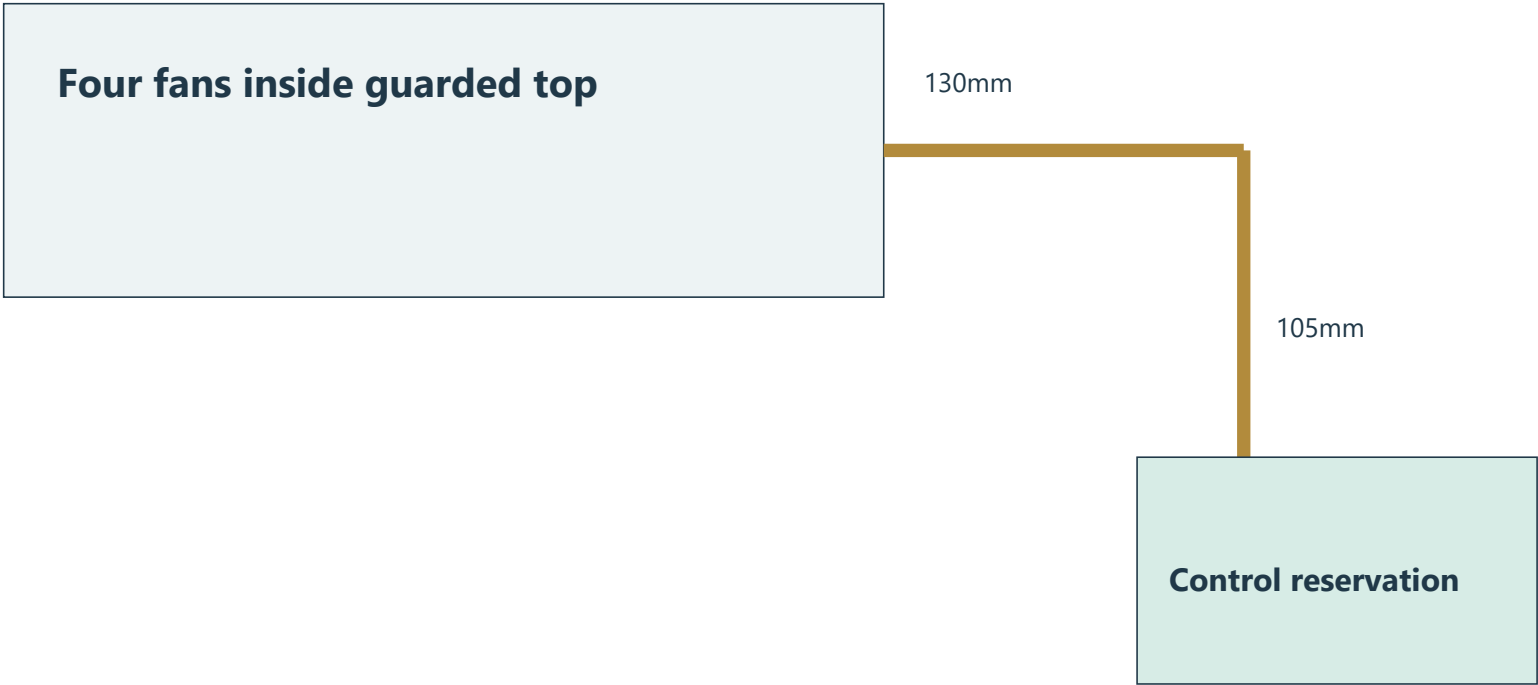
Feature	Coordinates / dimensions
Bracket	25 x25 x2 equal angle,30 long. Four per filter. Nominal sharp bend; actual extrusion root fillet and edge condition must fit.
Bracket x positions	Segments60-90,200-230,320-350,460-490. Fastener centres75,215,335,475.
Seat attachment	M4x20 proposal; dia4.5 at z37. Front and rear seat holes correspond. Washers/nut retention to be detailed.
Ledge attachment	M3x20 proposal through dia3.5 at y-15 (front) or375 (rear), at the four x centres. Head must be flush at/below z57.35.
Load path	Filter frame -> plywood ledge -> angle -> seat bolt/panel -> cabinet/base. Do not rely on filter clamping friction to carry weight.

Do not blindly countersink thin plywood: match the actual head, verify remaining material and check that the filter sits flat. The clamp studs, shelf fasteners and guard mounts were checked together in the same CAD.

Conditional load arithmetic - NOT a structural pass	Result
Guard bracket example:50N on one bracket,10mm arm;40mm wide x2mm strip. $\sigma = F L / (b t^2 /6)$.	18.75 MPa nominal
Shelf example:assumed2kg filter x2 handling factor, all load on one bracket,25mm arm;30mm wide x2mm strip.	49.05 MPa nominal
Missing before a strength judgement	Actual material allowable, bend/holes, connection bearing/pull-out, load distribution, guard flexure and abuse cases. These assumed loads are not compliance tests.

E02 Cable route and revised pressure

The electrical architecture is unchanged. This sheet reserves a route; it does not release a gland, wiring or restart protection.



Amber route: proposed12mm-diameter corridor. No12mm guard hole is authorized. Select a closed-entry gland/connector and cable restraint that preserve the guard. OEM fan leads may need extensions after internal routing and service loops are measured.

Release boundary

Detailed geometry now exists for the attachments. Exact products/materials, joint strength, guard manufacture/reach, seal compression/leakage, mass/stability, cable entry and restart protection still need closure. Physical prototypes built:0. No supplier contacted, purchase or installation made.

Actual work still needed
Confirm the third-party fan/controller chain, cable reach, connector retention, enclosure mounting and actual socket/plug arrangement. The control chain still lacks the required manual restart after power restoration. No safety function is assigned to the laptop or monitoring code.

Guard change	R01 assumption
Perforated field	300 x300mm inside a320mm face
Free area per face	0.036m2 assuming40% field porosity
Whole-face open fraction	35.15625%, previously40%
Calculated flows, low / middle / loaded	359.2 / 337.1 / 297.5 m3/h

These are screening scenarios, not measured ratings. The new guard border raises the calculated resistance; R00 airflow numbers no longer apply unchanged. Exact filter curve, guard loss coefficient and installation effects remain unknown. No CADR or outdoor coverage follows.